



SOLAR TRACKER MARKET

Data Visualization Application Development



Tracker Market Intelligence: a dashboard for the solar tracker market

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1. Motivation

This project comes out of my internship in strategy consulting. I am working on a project for a solar tracker manufacturer, and my part of the work is the market: how much solar is being installed in each country and segment, how much of it uses trackers, how large the retrofit opportunity is, and how solar and tracker prices differ across regions and why. Today all this sits in a large Excel model that is fed by hand from industry reports, databases and expert knowledge.

A solar tracker is a structure that rotates the panels so they follow the sun during the day. Compared with a fixed-tilt structure it produces roughly 15–25% more energy, which is why it has become the standard choice for large ground-mounted plants in the United States, the Middle East and Latin America. The market is growing quickly (134 GWdc of trackers were shipped worldwide in 2025, 19% more than in 2024, according to Wood Mackenzie), but where that growth happens is shifting: the US passed 40 GWdc for the first time, Saudi Arabia has overtaken Spain as the third-largest market, and Africa is growing fastest.

For a manufacturer, a country is not attractive just because it installs a lot of solar. Most residential and commercial systems sit on rooftops and never use a tracker, so what matters is the utility-scale segment, the share of it built with trackers, the installed base of older fixed-tilt plants that could be retrofitted, and the price a tracker can be sold at in that region. Those prices vary widely between regions, and the reasons behind the gaps (module prices, tariffs, steel, freight, labour, local content rules) shift over time. Deciding where to compete means looking at all of these together, region by region.

Providers such as Wood Mackenzie or BloombergNEF publish this kind of information, but their reports are expensive, come out once or twice a year and arrive as PDFs or spreadsheets rather than as a tool that can be filtered and compared. At the same time, much of the underlying data is public and updated often (installed capacity by country, new utility-scale plants, cost benchmarks, steel and freight prices, exchange rates), but it is scattered across many sources.

The contribution of this project is to bring all of that into one application that updates itself every day, measures the real size of the tracker market in each region (utility-scale additions times the attach rate, plus retrofit demand), explains why prices differ between regions, and lets the user change the main assumptions. On top of that I want to add a forecasting layer, so the application shows not only where each region has been but also where it is likely to go.

Because the project data is confidential, the manufacturer is never named, and the series that come from the internal model, such as attach rates or retrofit volumes by country, will be shown with illustrative values built from public figures. The application will accept an Excel or CSV upload with the real data, so anyone who has it can run the same analysis on it.

The goal, then, is to turn a manual and static market analysis into an interactive application that stays up to date on its own, compares regions on the same basis and helps decide which markets are worth competing in.

2. Strategic vision

The application covers the solar market outside China, which is where Western tracker manufacturers compete, and groups countries into four regions: North America, Europe, Latin America and Rest of World. Every chart can be read at country or regional level and split by segment: residential, commercial and industrial (C&I) and utility-scale, which is by far the largest and the one where trackers are actually used. The tool is not meant to replace the team's market model; it is a support tool that works alongside it, keeps it up to date and makes the day-to-day analysis easier.

The intended users are strategy and business development teams at tracker manufacturers, as well as consultants working on projects in the sector. The main benefit is time, since nobody has to look up and paste every new figure. Every region is also measured the same way, and the source of each number is visible.

The content follows the questions behind the market analysis: how much solar is being installed in each region and segment, how much of it is built with trackers, how large the retrofit opportunity is, why solar and tracker prices differ between regions, and how all of this is likely to evolve. These pieces combine into the size of the tracker market in GW and in value, from the historical data through to the forecast. The user will be able to filter by region and segment and adjust the main assumptions, such as those behind retrofit demand, and see the effect straight away. The data that is refreshed every day, such as steel and aluminium prices, exchange rates and solar generation, will not sit only in a separate real-time view. It will also be built into the rest of the analyses, so the user can see how a change in these variables affects tracker costs, prices, market size and the model as a whole. How this content is split into screens will be defined as the project develops.

As examples of use, a commercial director could pick a region and see in one place how much utility-scale solar is being built, what share of it uses trackers and how much extra demand could come from retrofits. They could also check why a tracker plant costs more in one region than in another, follow how a rise in steel or freight prices feeds into tracker costs, or compare where each region's tracker market has been and where it is heading.

3. Technical vision

The data comes from free public sources, split by how often they change. So far, three sources have been looked at for the daily automatic download, none of which needs an API key: the European Central Bank's reference exchange rates (euro against the US dollar, Brazilian real, Indian rupee and Chilean peso), Yahoo Finance through the yfinance library for US hot-rolled steel and aluminium futures and for listed solar and tracker equities, and Red Eléctrica's REDData API for daily solar PV generation and installed capacity in Spain. The rest are published monthly or yearly and are loaded when new figures come out: IRENA and Ember for installed capacity by country, the US Energy Information Administration for new utility-scale plants, IRENA cost reports and the NREL and Berkeley Lab benchmarks for system costs (including the comparison between tracker and fixed-tilt plants), and freight indices and module prices from the industry press. The internal market model in Excel adds the attach rates, the fixed-tilt installed base and the segment split, and it can be replaced by uploading a different file from the application. These are the sources explored so far; as the project develops, other sources and variables are likely to be added, both for the daily update and for the less frequent series.

A Python pipeline downloads the data, cleans it, aligns it by country, segment and date, and computes the derived metrics: tracker market size in GW and in value, the split between new build and retrofit, year-on-year growth and regional price indices. The daily series are not kept apart from the rest: they are linked to the model, so a change in, for example, steel prices or exchange rates feeds through to tracker cost and price estimates and from there to the value of the market. Everything is stored in a single table of observations (source, series, date and value) plus a log of every update. The initial plan is to use SQLite locally and PostgreSQL on AWS RDS in production; switching between them only means changing one connection setting. The first run downloads three years of history. After that, each daily run downloads the last two weeks and replaces those dates, which picks up any revisions and makes it safe to run the update more than once. If one source fails, the others are still updated and the error is recorded and shown in the application.

The forecasting layer will estimate PV additions, attach rates and prices by region for the coming years. Because the series are annual and fairly short, one model per country makes little sense, so a single model will be trained on all countries together, using past values and variables such as module prices or recent

growth. The idea is to test options such as a regularised linear regression and a tree-based model and compare them against a very simple baseline that extends the trend of recent years, to check whether they actually improve on it. In the application, the forecasts will be shown together with the historical data but clearly separated from it.

Everything will be presented in an application built with Dash and Plotly. The region and segment filters and the assumption controls will update the results without reloading the page. The code is written in Python and hosted on GitHub. In the current version, the daily update runs in two places: a scheduled GitHub Actions job every morning, which also runs the pytest test suite before downloading, and a scheduler inside the application (APScheduler) that updates the data at 06:00 Madrid time and catches up on start-up if the last update is more than 20 hours old. Finally, the application will be deployed on an AWS EC2 instance, with the database on RDS. A first working version already exists: the daily data is live, while the market model series use illustrative values until the real data is uploaded.